Real-Time Video Inference on 8× NVIDIA RTX PRO 5000

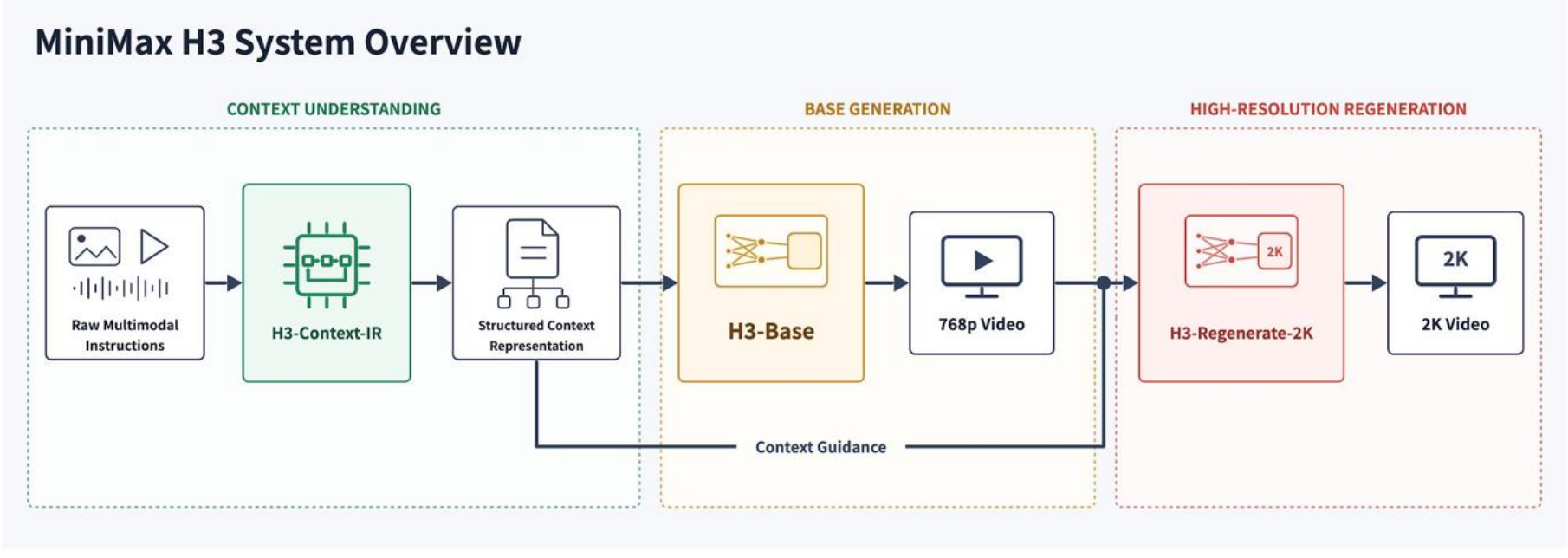
MiniMax H3 with vLLM-Omni

Shunyang Li | Public Technical Brief | October 2026



Real-Time Video Generation Is a Distributed-Systems Problem

For each request, the real-time target is end-to-end latency within the clip's playback window; across the evaluated 60K–1M-token range, shape changes alter memory demand, kernel geometry, communication volume, and CUDA Graph reuse.



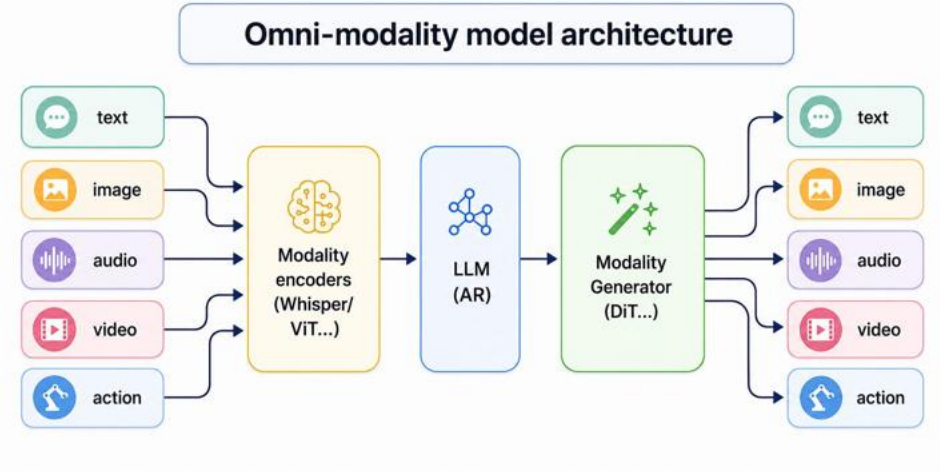
EVALUATED PATH · 8 DiT STEPS × 50 BLOCKS = 400 REPEATED BLOCK EXECUTIONS

Under SP8, the evaluated path repeatedly couples compute with Q/K/V and Gate/O exchange; launch gaps, transport fences, rank skew, and the decoder tail accumulate on the critical path, while the optimized system must preserve numerical equivalence to the reference path under the defined tolerance.

Sources: vLLM-Omni v0.26 public deck, slide 23 · author evaluated workload envelope and eight-GPU execution path

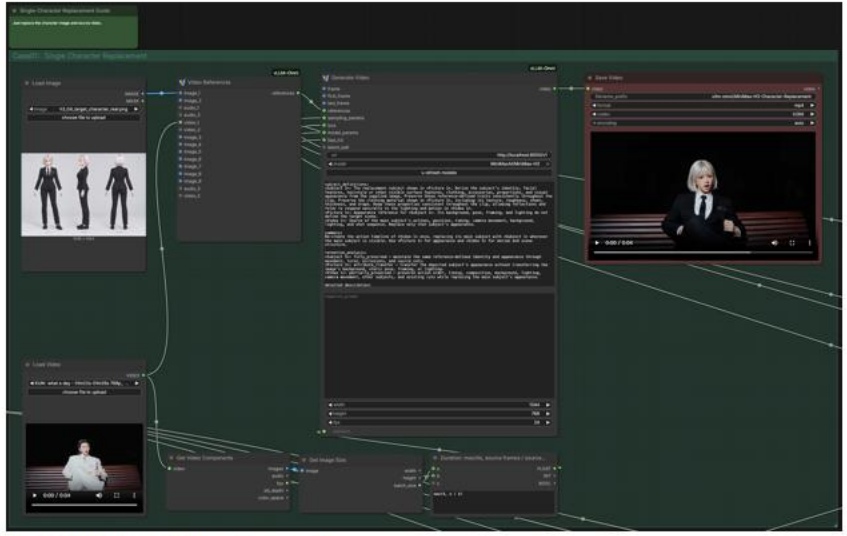
vLLM-Omni Connects Visual Workflows to Multi-GPU Serving

ComfyUI remains the authoring surface; vLLM-Omni owns serving, stage placement, batching, and multi-GPU execution



OFFICIAL vLLM-OMNI ARCHITECTURE

One serving layer spans omni-modal inputs, autoregressive reasoning, and DiT generation.



COMFYUI WORKFLOW · PR #9194

MiniMax-H3 character replacement calls the vLLM-Omni Generate Video endpoint; PR #9194 remains open.

Sources: vLLM-Omni v0.26 public deck, slide 22 · vLLM-Omni PR #9194 (open)

Frontier Models · The Cards

One Blackwell architecture (CC 12.0/12.1) spans several very different deployment envelopes

SKU	CC / SMs	Memory	Power	Interconnect
RTX PRO 5000 Blackwell 8-GPU baseline	12.0 / 110	72 GB GDDR7 ECC 48 GB variant	**350 W configured*** **300 W public TBP**	PCIe Gen5 ×16 no GPU NVLink
RTX PRO 6000 Blackwell WS / Server	12.0 / 188	96 GB GDDR7 ECC	600 W	PCIe Gen5 ×16 no GPU NVLink
GeForce RTX 5090	12.0 / 170	32 GB GDDR7	575 W	PCIe Gen5 ×16 NVLink: no
DGX Spark · GB10	12.1 / 48	128 GB unified LPDDR5X	140 W SoC 240 W system	NVLink-C2C in SoC 200 Gb/s ConnectX-7

SHARED BY DISCRETE CARDS · BLACKWELL FP4 · GDDR7 · PCIe GEN5 · NO GPU-TO-GPU NVLINK

Watch sustained clocks, not peak TOPS: controlled tensor-heavy tests reported SwPowerCap in 94.8–100% of valid samples; one rank stayed at 1.942 GHz and ran the same fine-attention input 6.78% slower. A 600 W-class design offers more headroom, but still requires a matched chassis A/B.

*350 W is the local configured/enforced limit; NVIDIA publicly specifies 300 W TBP for RTX PRO 5000.

Sources: NVIDIA product specifications · CUDA GPU Compute Capability · local system inventory and controlled clock study